

VitalFlw: Project Specification and Operational Blueprint

1. System Architecture & Technical Stack

The platform is designed as a multi-tenant SaaS application catering to small and mid-sized clinics. The architecture prioritizes data isolation, rapid deployment, and high-performance user interfaces.

Layer	Technology	Reasoning
Frontend	Next.js 14+ (App Router)	Marketing site and high-performance dashboard unified in one repository.
Styling	Tailwind CSS + shadcn/ui	Rapid UI development maintaining medical-grade accessibility standards.
Backend	Node.js / Server Actions	Type-safe communication between the user interface and database layer.
Database	PostgreSQL (via Supabase)	Relational integrity combined with strict Row Level Security (RLS).
ORM	Prisma	Type-safe database querying and streamlined schema migrations.

Layer	Technology	Reasoning
Auth & Roles	Clerk	Native "Organizations" support allowing robust clinic multi-tenancy.
CMS (Blog)	Sanity.io or Contentful	Headless CMS optimized for the Content Writer to manage SEO efforts.

2. Role-Based Access Control (RBAC)

To scale VitalFlw effectively, the system distinguishes between Platform Roles (internal business management) and Clinic Roles (tenant medical practice management).

2.1 Platform Roles (Internal)

- **Super Admin:** Full system access, cross-clinic subscription management, platform revenue analytics, and account suspension capabilities.
- **Content Writer:** Restricted access isolated to the Headless CMS for publishing SEO-optimized marketing and healthcare content. Strict policy of zero access to application databases or clinic records.

2.2 Clinic Roles (External/Tenants)

- **Clinic Admin:** Organization owner. Manages clinic-specific settings, staff invitations, subscription billing, and overarching clinic metrics.
- **Doctor:** Clinical scope. Access to assigned patient histories, digital prescription generation, and individual consultation schedules.
- **Receptionist:** Operational scope. Access to appointment scheduling, patient onboarding, front-desk queue management, and invoicing.
- **Nurse/Pharmacist:** Support scope. Access to vital sign logging, inventory tracking, and pharmaceutical dispensing.

3. Module Specifications

3.1 Scheduling & Queue Management

- **Functionality:** Calendar-based appointment booking, drag-and-drop rescheduling, and real-time status tracking (Waiting, In-Consultation, Completed).
- **Data Structure:** Links Patient IDs to Doctor IDs with timestamp constraints.

3.2 Patient Records (EMR)

- **Functionality:** Centralized digital medical history, file upload capabilities for radiological scans, and secure demographic data storage compliant with the DPDP Act 2023.
- **Data Structure:** Master patient index isolated securely by clinicId.

3.3 Billing & Invoicing

- **Functionality:** Automated GST-compliant invoice generation and payment tracking. Adaptable line-item additions for consultations, procedures, and pharmaceutical sales.
- **Data Structure:** Transaction ledgers tied to patient and clinic identifiers.

3.4 Staff & Human Resources

- **Functionality:** Role assignment, shift scheduling, and access revocation.
- **Data Structure:** User mapping to Clerk Organization roles.

3.5 Digital Prescriptions

- **Functionality:** Digital entry of medical advice, dosage drop-downs, and one-click PDF generation or secure digital sharing via WhatsApp.
- **Data Structure:** Immutable prescription records linked to specific consultation sessions.

3.6 Reports & Analytics

- **Functionality:** Business intelligence dashboards displaying daily revenue, patient volume, and service popularity.
- **Data Structure:** Aggregated read-only queries on billing and scheduling tables.

3.7 Notifications & Engagement

- **Functionality:** Automated reminders for upcoming appointments to reduce no-show rates.
- **Data Structure:** Webhook triggers integrated with third-party messaging APIs.

3.8 Inventory Management

- **Functionality:** Tracking of pharmaceutical stock, low-stock alerts, and expiry warnings.
- **Data Structure:** Inventory tables with decrement triggers linked to the billing module.

4. Operational Pricing & Cost Estimation

The following estimates outline the monthly operational costs to run the infrastructure in Indian Rupees (INR). The startup phase is heavily optimized to leverage free tiers during development and initial user acquisition.

Service Layer	Provider	Development / MVP Cost	Production Cost (Scale)
Domain & Workspace	Google Workspace	~₹150 / user / month	Scales per employee
Hosting & Compute	Vercel	₹0 (Hobby Tier)	~₹1,650 / month (Pro Tier)
Database & Storage	Supabase (PostgreSQL)	₹0 (Free Tier)	~₹2,000 / month (Pro Tier)
Authentication	Clerk	₹0 (Up to 10k MAU)	~₹2,400 / month (Pro Tier)
Communications	MSG91 / Twilio	Pay-as-you-go	Volume Dependent
Content Management	Sanity.io	₹0 (Free Tier)	₹0 (Often sufficient)

Phase-based Total Estimates

- **Development Phase (Months 0-3):** ~₹150 / month (Primarily the Google Workspace setup).
- **Initial Production (Months 3-8, Early Adopters):** ~₹1,200 - ₹2,000 / month (Workspace + usage-based notification API limits).
- **Scaling Phase (15+ Clinics):** ~₹7,000 - ₹8,200 / month (Transitioning Vercel, Supabase, and Clerk to Pro tiers to support production-level traffic and storage requirements).